

TASK 03 REPORT

INTERNSHIP
CYBERSECURITY

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WEEK 3 TASK REPORT

Internship Cybersecurity 2026 Project: Advanced Security and Final Reporting

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WEEK 3: ADVANCED SECURITY AND FINAL REPORTING

1. BASIC PENETRATION TESTING

In the final week, I performed controlled penetration testing to simulate real-world attack scenarios. These tests evaluated the resilience of the application following the Week 2 remediations.

- **Vulnerability Validation:** Confirmed that critical exploits (SQLi, XSS) identified in Week 1 are no longer executable.
- **Access Control Verification:** Verified that the JWT-based authorization successfully prevents horizontal and vertical privilege escalation.
- **Resilience Summary:** The application successfully resisted all basic penetration attempts, indicating a significantly hardened security posture.

2. LOGGING AND MONITORING

To improve incident response capabilities, I integrated a comprehensive logging framework:

- **Event Tracking:** All login attempts, administrative actions, and failed authorization attempts are now recorded.
- **Security Logs:** Logs are saved to a dedicated `security.log` file, enabling offline audits and monitoring of unusual access patterns.

- **Visibility:** Enhanced visibility allows for real-time detection of brute-force or unauthorized traversal attempts.

3. SECURITY BEST PRACTICES CHECKLIST

A final security checklist was established to maintain the application's integrity post-deployment:

- [x] **Input Validation:** All user data is sanitized via the `validator` library.
- [x] **Secure Hashing:** Passwords are never stored in plain text; Bcrypt is used.
- [x] **Auth Protection:** APIs are secured with stateless JWT authentication.
- [x] **Header Security:** Helmet is applied to enforce CSP and prevent clickjacking.
- [x] **Encrypted Transit:** HTTPS is enabled for all client-server communications.
- [x] **Audit Trail:** Continuous logging is active for all critical system events.

FINAL SUMMARY

The three-week internship successfully progressed from **Vulnerability Identification** (Week 1) to **Active Remediation** (Week 2), and finally to **Security Verification and Monitoring** (Week 3). The application now adheres to industry-standard security protocols, minimizing the risk of data breaches and unauthorized access.